



TITLE: AMAZON ROBOTICS LIGHTING SPECIFICATION		
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AMAZON ROBOTICS LLC

300 RIVERPARK DRIVE

NORTH READING, MA 01864

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REVISION HISTORY

Rev.	Description of Change	Sections Affected	Originator/Department
01	New document release	ALL	B. Klein - Engineering
02	Removed all references to Nike and Non-Nike stations changed to “stations that require a dark zone” and “stations that do not require a dark zone” Added Dark Zone, X-DU, SHALL, & SHOULD definitions Added AR-ID to the list of definitions Added new section 1.4 to define requirement types. This is meant to clarify what requirements the construction team needs to adhere to as what requirements the AR hardware engineering team needs to adhere to. Updated entire document to indicate requirements that are construction and requirements that are for hardware engineering. Updated section to indicate that there are stations that do and do not require a dark zone; references were added to their respective sections. Also, a dark zone definition was added for clarification purposes. Added a new table 1 showing the stations that don't require a “dark zone” Added clarification as to what equipment drives the need for a “dark zone” and added table 2 for stations that require the “dark zone” Added new figure 2 showing a side diagram of the robotic delivery station (RDS). Updated section 2.1.2 so the construction requirements, related to the dark zone, show up first and engineering requirements show up second. This is to avoid confusion for construction stakeholders consuming this document. Updated section to include a tolerance on the dark zone requirements based on a request from AR Deployment engineering via SIM https://issues.amazon.com/issues/ARSpec-7	ALL 1.2 1.3 1.4 ALL 2.1 2.1.1 2.1.2 2.1.2 2.1.2 2.1.2.1	B. Klein - Engineering

	Added clarification on what the purpose of the “Dark Zone” is to help the construction teams; this is based on numerous questions fielded related to the REV 01 release. Added requirement for dimmers on fixtures around the dark zone to maximize flexibility. Added new figure 3 to clarify allowable illuminance and tolerances in the dark zone. This shows two subzones known as the Outer Dark Zone and Inner Dark Zone Added new table 1 to help clarify allowable illumination requirements as you move away from the AR perimeter fence. Added verbiage on ramp-up illuminance from the dark zone to the end of the station perimeter. Added a note to clarify that the lighting engineer should determine method of achieving the dark zone. Swapped sections 2.1.2.1 and 2.1.2.2 to flow better. Updated diagram to make it clear where the pod face location is. Updated maintenance cage requirement for dimmers from a “shall” to a “should”. Indicated that using dimmers in the maintenance cage is the recommended implementation. Added X-DU to the table and placed a footnote to indicate that there are passive and active versions of the camera. The active version has an illumination board operating at a wavelength of 850nm.	2.1.2.1 2.1.2.1 2.1.2.2 2.1.2.1, 2.1.2.2 2.1.2.3 2.2 3.1	
03	Updated Figures 3 & 4. Figure 3 labels changed from Nike and Non-Nike stations to “stations that require a dark zone” and “stations that do not require a dark zone”	2.1.2.1, 2.1.2.2	M. DellaCroce - Engineering
04	Added DTP, GoCart, ToteASRS, T-DU, definitions and updated others as needed Added T-DU to list of acronyms Further clarified Construction vs. Amazon Robotics Hardware Engineering requirements to (building equipped lighting vs station equipped lighting respectively) Removed: Pick-to-Rebin (P2R) Added: ToteASRS Consolidation, ToteASRS Universal, ToteASRS ARSAW, ToteASRS Pick-to-Rebin, Cardinal, and Sparrow to Table 1; Stations that do not require a dark zone.	1.2 1.3 1.4 2.1.1	M. DellaCroce - Engineering

	Added: Pick-to-Rebin (P2R) and Softlines Micro-Fulfillment Universal to Table 2; Stations that do require a dark zone. Added: T-DU to obstacle detection system wavelengths to list of AR equipment infrared light sources table. Updated Figure 10 to include all 12 light measuring locations at pod face/station. Rounded Dark zone measurements from 1219mm to 1220mm for ease of site validation, and 1238mm to 1240mm throughout Pod face light level measurements and station ambient measurements are clarified as Hardware engineering requirements and dark zone document sections are clarified to be validated by construction as they are to be provided by construction. Clarified that ambient building lighting shall also apply to the highways within the pod farm.	2.1.2 3.1 Figure 10 All All All	
05	Added Transfer Cartesian Cell and ToteASRS Stations; SAUS, SAPS, and P2R to stations that do not require dark zones. Added Proteus definition. device wavelengths, and glare interference requirements throughout.	2.1.1 All	M. DellaCroce - Engineering
06	Added SRBRS, MSC, FSS, and TCC / TCIS acronyms Added ToteASRS TCC / TCIS specialized dark zone requirements Added TCC / TCIS specialized dark zone validation requirements for tech deployment Removed AR-ID from Dark Zone requirement as AR-ID does not require Dark Zones, added Scanless Decant station to non-dark zone required stations	1.3 2.1.3 5.1.4 All	M. DellaCroce - Engineering
07	Added Aether to definitions Clarified wording for ToteASRS stations lighting referenced in relation to the TCC's lighting requirements.	1.2 2.1.3	M. DellaCroce - Engineering

	Added new Chapter for Aether building lighting requirements and Aether panel compatibility and reference to ARPN #811-00252	4.0	
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1 Introduction

The purpose of this document is to provide lighting requirements for buildings that contain AR enabled systems. Lighting requirements include ambient lighting, sunlight and allowable infrared light based on building and / or drive unit type. This document also includes methods for performing illuminance and IR radiation measurements.

1.1 Site Applicability

The requirements contained herein apply to all AR enabled buildings.

1.2 Definitions

- **Dark Zone** – The area around the fence line between the AR field and the station interface that is kept dark (<5FC) as a means to control lighting when the station is deployed with an integrated task light.
- **Drive Unit** – An autonomous robot used in the Amazon Robotics Mobile Fulfillment System to pick up and move pods.
- **DTP** – Dense Tote Pod. Mobile shelving unit used in the largest Amazon Robotics Mobile Fulfillment System to store inventory.
- **GoCart** – Container to transport Amazon packages in the outbound dock area for outbound shipping.
- **Pod** – Mobile shelving unit used in the Amazon Robotics Mobile Fulfillment System to store inventory.
- **RSP** – Robotic Storage Platform. Multi-level platform or mezzanine to support multiple floors of the Amazon Robotics Mobile Fulfillment System in the same building.
- **ToteASRS** – Tote Automated Storage and Retrieval System which utilizes the Titan drives and DTPs.
- **G-DU Utilization** – The smaller Amazon Robotics Mobile Fulfillment System which utilized the G drives and pods.
- **S-DU Utilization** – The larger Amazon Robotics Mobile Fulfillment system which utilized the S drives and pods.
- **H-DU Utilization** – The successor to the G system which utilizes the H drives and H pods.
- **T-DU Utilization** – The largest Amazon Robotics Mobile Fulfillment System which utilizes Titan drives and DTPs.
- **P-DU Utilization** – The Amazon Robotics Fulfillment system, which utilizes Pegasus, Pegasus-Tall and Pegasus-Large drive units and moves individual packages around a sort center.
- **P-L-DU Utilization** – The Amazon Robotics Fulfillment system, which utilizes Pegasus-Large drive units and moves individual packages around a sort center.
- **Proteus DU** - Autonomous Drive Unit to serve in the Outbound Dock area of Amazon buildings. Proteus will autonomously move GoCarts throughout the Outbound Dock.
- **P-T-DU Utilization** – The Amazon Robotics Fulfillment system, which utilizes Pegasus-Tall drive units and moves individual packages around a sort center.
- **SHALL** – Denotes a hard requirement

- **SHOULD** – Denotes a recommendation, but not a hard requirement
- **X-DU Utilization** – Mobile robotic platform that accepts various payloads (a.k.a. Xanthus).
- **Task Lighting** – Lighting that is integrated into the station for station illuminance.
- **Ambient Lighting** – Lighting that is present in the environment caused by building infrastructure lighting and / or sunlight. Task lighting is excluded from ambient lighting.
- **Aether** – AR Lighting panel that controls the AR field lighting on a single switch per floor for AR fulfillment systems.
- **Min / Max Ratio** – The ratio of illuminance in the associate work area between the average maximum light and average minimum light measured required to maintain associate experience; this is 4:1.

1.3 Acronyms

- **AFF** – Above the finished floor
- **AR** – Amazon Robotics
- **AR-ID** – Enhanced sensor system used for scanning barcodes
- **DU** – Drive Unit
- **FC** – Foot-Candle unit of illumination (one lumen per square foot or 10.764 lux)
- **FIDO** – Field Intrusion Detection
- **FSS** – Floor Safety System
- **G-DU** – G-Drive Unit
- **IDS** – Indent Detection System
- **IR** - Infrared
- **H-DU** – Hercules Drive Unit
- **lx** – Lux, SI unit of illuminance (one lx is equal to 0.0929 FC)
- **mm** – Millimeters
- **MSC** – Main Safety Controller
- **nm** – Nanometer (in this document, defines the wavelength of light)
- **RDS** – Robotic Delivery Station
- **RGB** – Red, Green, Blue
- **P-DU** – Pegasus Drive Unit
- **P-L-DU** – Pegasus Large Drive Unit
- **Proteus** – Proteus Drive Unit
- **P-T-DU** – Pegasus Tall Drive Unit
- **S-DU** – S-Drive Unit
- **SRBRS** – Short Range Broadcast Radio Frequency System
- **TCC / TCIS** – Transfer Cartesian Cell / Transfer Cartesian Induct Cell
- **T-DU** –Titan Drive Unit
- **X-DU** – X-Drive Unit

1.4 Requirement Types

Requirements within this specification fall into two types of requirement categories; “Construction requirements” and “Engineering requirements”. Each requirement will be identified with the requirement type to avoid confusion when interpreting the specification.

- **Construction Requirements** – These are requirements that the construction team needs to meet at the facility / building (building equipped lighting).
- **Amazon Robotics Hardware Engineering Requirements** – These are requirements Amazon Robotics hardware engineering needs to meet for the station implementation (station equipped lighting).

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2 General Requirements

The following sections are broken down into AR Stations lighting requirements in Sections 2.1 – 2.1.2.4 (Fabric Fulfillment Stations which all require dark zones and ToteASRS Fulfillment Stations which all do not require dark zones). ToteASRS Specific TCC / TCIS lighting requirements found in Sections 2.1.3 – 2.1.3.2 which are in-field (internal AR field) specialized dark zone lighting requirements only, and do not impact the associate AR Stations outside of the fence line.

2.1 Lighting Requirements – Stations

In this document, Stations are identified as ***Stations that do not require the “Dark Zone”*** (section 2.1.1 table 1) and ***Stations that require the “Dark Zone”*** (section 2.1.2 table 2). See the respective sections for details regarding station “dark zone” applicability.

- **Note:** The purpose of the “dark zone” is to ensure the area where the station is to be deployed is sufficiently dark such that when the station is deployed, the ambient lighting conditions don’t add too much light to the light provided by the station task light when turned on. If the ambient conditions are too bright prior to station deployment, this may cause improper station operation that may require remediation.

2.1.1 Stations that do not require the “Dark Zone”

Stations that do not require the “dark zone”, per this document, are defined as stations that do not have features like the Intent Detection System (IDS) or Bin Illumination. Table 1 provides station names / ID’s that are considered exempt from the “dark zone” requirements.

Table 1 - Stations that do not require a “Dark Zone”

Station Name	Station ID on Composite (if different)
ARTEMIS	ARTEMIS
ARXD Cross-dock/Cart Sort	UNIV
ToteASRS Semi-Automated Universal Station (SAUS)	ToteASRS SAUS
ToteASRS Semi-Automated Pick Station (SAPS)	ToteASRS SAPS
ToteASRS Pick-to-Rebin	ToteASRS P2R
ROBIN	ROBIN
CARDINAL	CARDINAL
SPARROW	SPARROW
Scanless Decant	DECANT

2.1.1.1- Stations without Dark Zone-Ambient Lighting-Associate Work Area

CONSTRUCTION REQUIREMENTS: The ambient lighting within the stations associate work area **shall** be between 27 foot-candles (290 lx) and 35 foot-candles (377 lx) following average maintained light levels measured 915mm above the finished floor (AFF) with max to min values not to exceed 4:1. This requirement **shall** be achieved regardless whether the building has windows, skylights or other light sources. It is recommended that the lighting **should** be on dimmers to maximize flexibility and ability to meet the current and future retrofit requirements.

- **Note:** For stations that contain platforms, measure with respect to the top of the platform. See section 5.1 for instructions on how to perform these measurements.
- **Note:** There are no “dark zone” requirements stations listed in table 1.

2.1.2 Fabric Fulfillment RSP levels Stations that require a “Dark Zone”

Below are the illuminance requirements to enable proper operation of the stations that require a “dark zone”. If the station has the Intent detection system (IDS) or the BIN illumination project, then the “dark zone” applies. See table 2 for a list of applicable stations that require the “dark zone”.

- **Note:** The illuminance requirements for the pod face are different from the actual associate work area.
- **Note:** Table 2 provides station names / ID’s that are considered stations that **require the “dark zone” (see section 2.1.2.1) for proper operation.**

Table 2 - Stations that require a “Dark Zone”

Station Name	Station ID on Composite (if different)
Universal (Non-ARXD Cross-dock sites)	UNIV
ARSTOW	ARSTOW
ARSAW	ARSAW
Robotic Delivery Station (RDS)	INDUCT or ICQA
Fabric Pod Pick-to-Rebin	Fabric Pod P2R

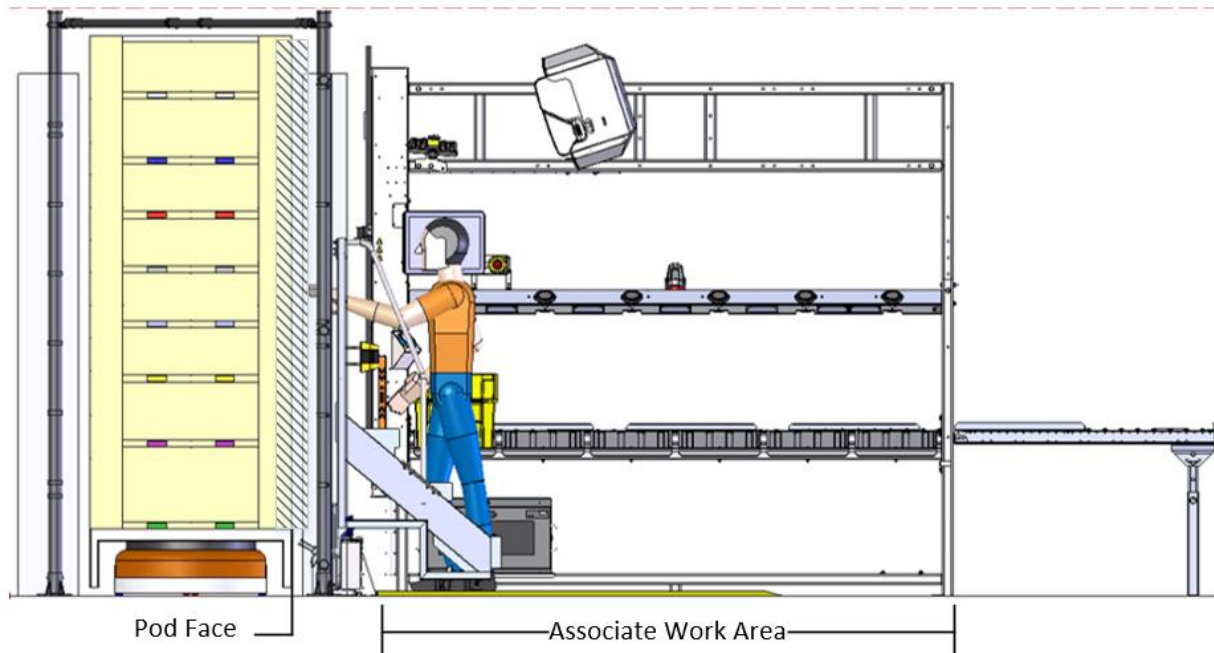


Figure 1 – Sortable FC Universal Station - Pod Face & Associate Work Area

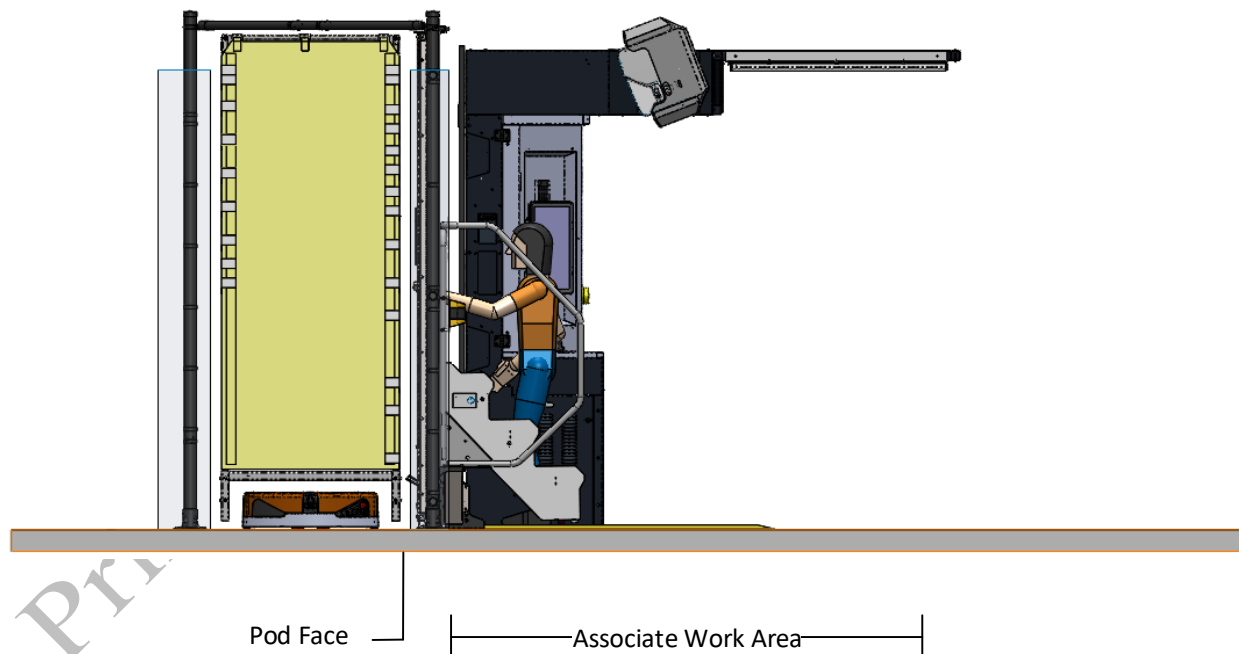


Figure 2 – Robotic Delivery Station - Pod Face & Associate Work Area

2.1.2.1 “Dark Zone” Location FABRIC FULFILLMENT ONLY

CONSTRUCTION REQUIREMENTS: With respect to the fence line, the dark zone shall be 1220 mm inside the AR field and 1220 mm outside the AR field, and extend the full width of the station. Figure 4 is for reference purposes only and is not to scale. The

dark zone should ramp / gradually increase to the full 27-35 foot-candles (290 - 377 lx) ambient building lighting when greater than 2440 mm from the perimeter face/pod face; see table 1 for details. This requirement shall also apply to the highways within the pod farm outside the dark zone area such that highways contain the ambient building lighting level requirements herein. Dark zone validation is a construction requirement.

- **Note:** Figure 3 is meant to help clarify allowable illuminance / tolerance within the Dark Zone from 2.1.2.1 - The most critical area for dark zone lighting is the vertical plane where the pod face (see figure 1 and 2 for locations) is presented to the Amazon associate at the fence line (Inner Dark Zone). The desire is to ensure this Inner Dark Zone area does not exceed 5 foot-candles. It is understood that some light overspill may occur towards the far ends of the dark zone (Outer Dark Zone) as 1220 mm is approached away from the fence line, therefore a tolerance was added to accommodate that.
- **Note:** How the dark zone is to be achieved is not specified in this document, and should be determined by the lighting engineer. However, the most effective method observed is to achieve the dark zone requirements and add remediation lights to under-illuminated areas around the dark zone/within the building.

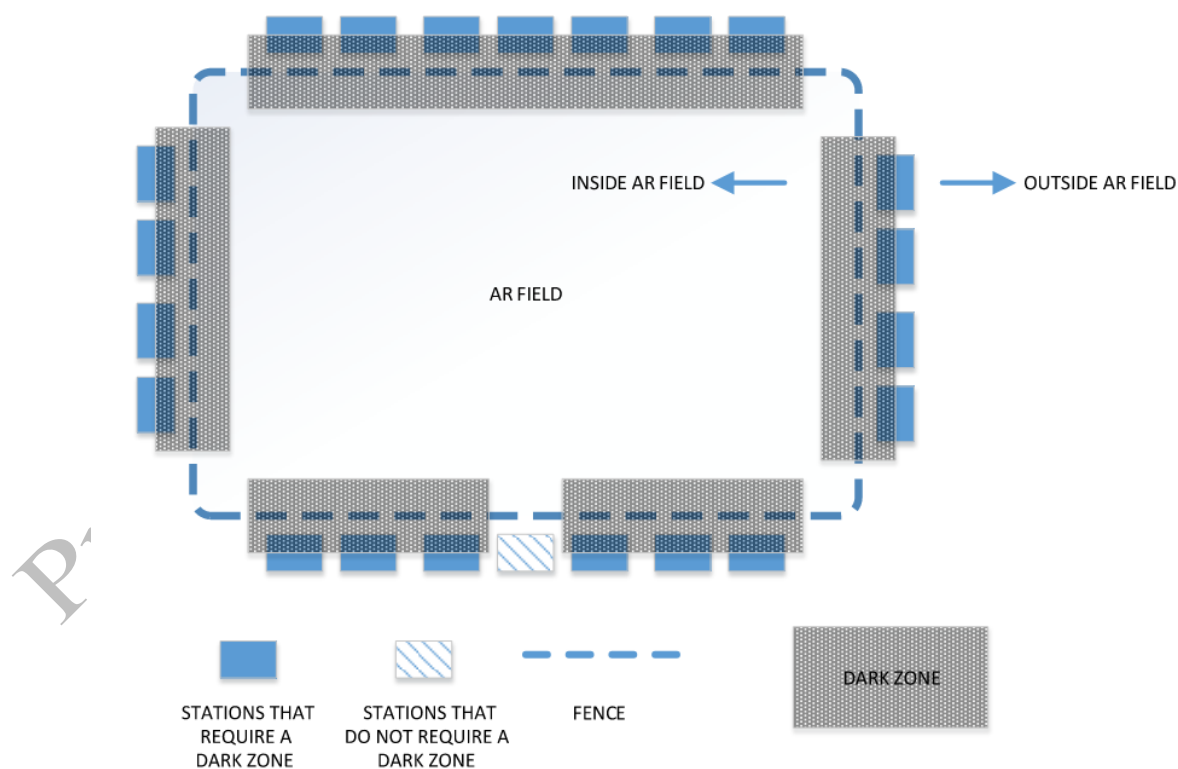


Figure 3 – Keep-Out Zone & Dark Zone Locations

2.1.2.2 “Dark Zone” Lighting FABRIC FULFILLMENT ONLY

CONSTRUCTION REQUIREMENTS: The lighting in the Dark Zone has two subzones: *Inner Dark Zone* and *Outer Dark Zone* (See Figure 3). The *Inner Dark Zone* **shall** not exceed 5 foot-candles or 50 lx as in figure 3. The *Outer Dark Zone* **shall** be 5 foot-candles (+3/-5 foot-candles) or 50 lx (+32/-50 lx) when a station is not installed as in Figure 3. The lighting around the dark zone **should** be on dimmers to maximize flexibility and ability to meet the specification requirements. Dark zone validation is a construction requirement.

- **Note:** Further details / clarification on these zones can also be found in Table 1.
- **Note:** The station that require the “dark zone”, as listed in table 2, integrates task lighting that will provide sufficient lighting to meet the requirements in section 2.1.2.3 and 2.1.2.4.
- **Note:** The purpose of the “dark zone”, is to ensure the area where the station is to be deployed is sufficiently dark enough such that when the station is deployed, that the ambient lighting conditions don’t add too much light to the light provided by the task light when turned on. If the ambient conditions are too bright prior to station deployment, this may cause improper station operation that may require remediation.

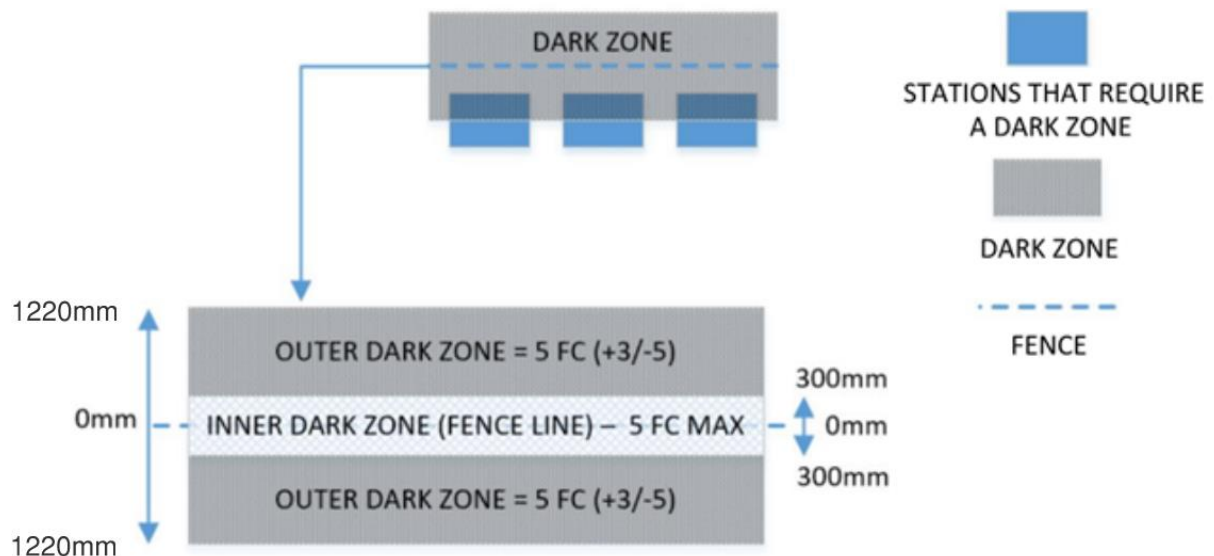


Figure 4 – Dark Zone (Inner Dark Zone and Outer Dark Zone Details)

Table 3 - Illumination by Distance from AR Perimeter Fence Table

Distance from AR perimeter fence (AR pod face)	Required illumination	Description
0mm to 300mm Inner Dark Zone	5 foot-candles MAX (50 lx)	This is the Inner Dark Zone and needs to be 5 foot-candles or less. This is required on the AR pod face for proper station functionality.
300mm-1220mm Outer Dark Zone	0-8 foot-candles (0 – 86 lx)	This is the Outer Dark Zone. The station task light will bring the work area illumination into the desired range referenced in section 2.1.2.4.
1220mm – 2440mm	<u>Outer Dark Zone</u> value up to 35 foot-candles (377 lx) - or target building illumination	Area allowed for ramp-up from dark zone to ambient building lighting specification.
2440mm+	27-35 foot-candles (290 - 377 lx) - or target building illumination	Ambient building lighting.

2.1.2.3 Ambient Lighting – Pod Face FABRIC FULFILLMENT ONLY

HARDWARE ENGINEERING REQUIREMENT: The ambient lighting on the pod face **shall not** exceed 15 foot-candles (161 lx). See figure 1 for location of the pod face. See section 5.1 for instructions on how to perform these measurements. See figure 5 for pod face location.

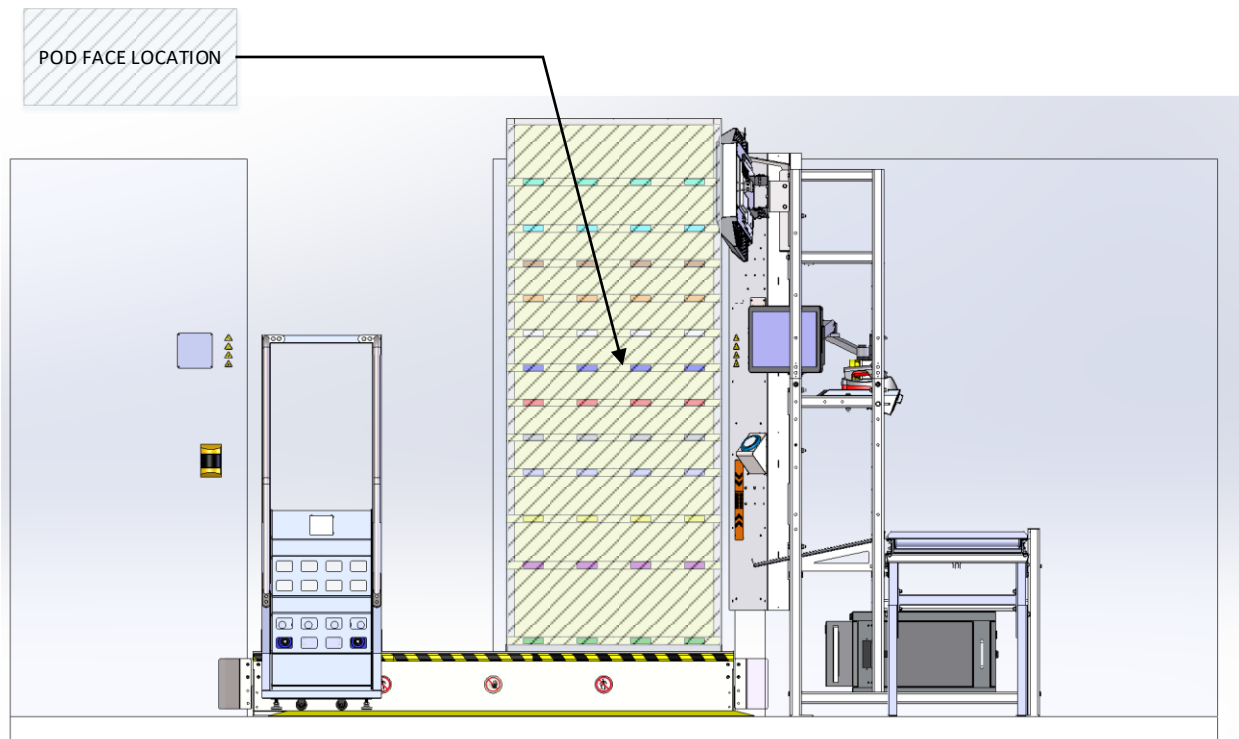


Figure 5 – Pod Face

2.1.2.4 Ambient Lighting – Associate Work Area

HARDWARE ENGINEERING REQUIREMENT: The ambient lighting within the stations associate work area **shall** be between 27 foot-candles (290 lx) and 35 foot-candles (377 lx) following average maintained light levels measured 915mm above the finished floor (AFF) with max to min values not to exceed 4:1. This requirement **shall** be achieved regardless whether the building has windows, skylights or other light sources.

Note: For stations that contain platforms, measure with respect to the top of the platform. See figure 1 for the location of the associated work area. See section 5.1 for instructions on how to perform these measurements.

2.1.3 ToteASRS Fulfillment RSP levels “Specialized Dark Zone” Requirement

Below are the illuminance requirements to enable proper operation of the TCC / TCIS as part of the ToteASRS fulfillment system. This specialized dark zone requirement applies to the robotics side of the AR field and not to the ToteASRS AR stations around the AR field on the pedestrian side of the fence line. ToteASRS stations do not require a dark zone.

2.1.3.1 “Specialized Dark Zone” Location

CONSTRUCTION REQUIREMENTS: With respect to the finished floor, toward the ceiling, inside the AR fence line, the specialized dark zone **shall** be 2650mm measured from the finished floor toward the ceiling. See figure 6 for a visual elevation representation of the specialized dark zone shown in gray in the AR field not to scale.

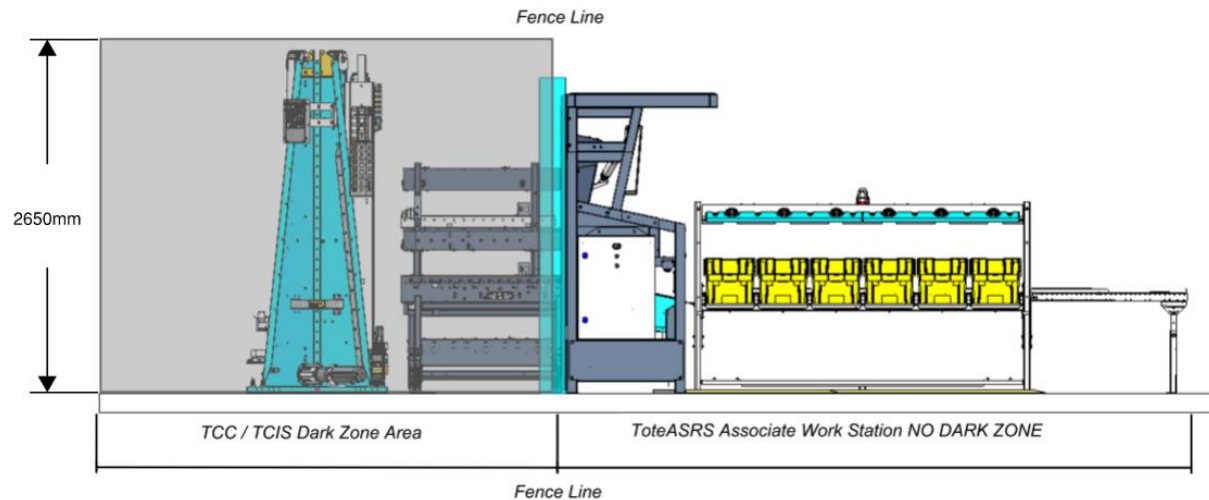


Figure 6 – TCC Dark Zone Area Divided by Fence Line to ToteASRS Station Area Elevation View

With respect to the fence line, toward the TCC and the AR field side of the fence, the specialized dark zone **shall** be 4000 mm inside the AR field and extend the full width of the TCC / TCIS in plan view (4520 mm). Use the TCC / TCIS anchorage survey points as a reference for location of the TCC / TCIS. See figure 7 for visual plan view representation not to scale.

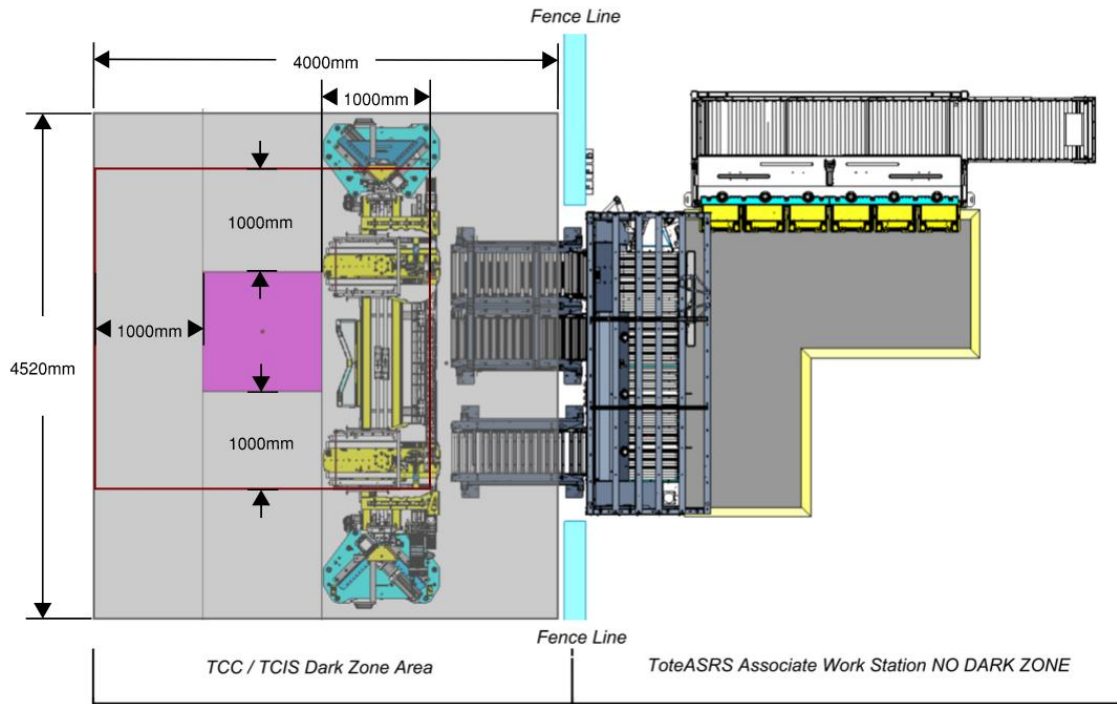


Figure 7 – TCC Dark Zone Area Divided by Fence Line to ToteASRS Station Area Plan View

The specialized dark zone space shall not exceed 35-foot candles (377 lx) of light anywhere in the specialized dark zone both in plan and elevation directions. The minimum specialized dark zone lighting shall not contain less than 5-foot candles (50 lx) of light. Dark zone validation is a construction requirement.

If light fixtures are present in the area of the specialized dark zone for TCC / TCIS, the fixtures shall be turned off to meet this performance requirement. Lighting within the specialized dark zone is permitted to range anywhere from 5-foot candles (50 lx) up to 35-foot candles (377 lx).

Ceiling light fixtures shall be physically located at least 1000mm away from the edges of the Hercules Dense Tote Pod nominal central cell shown in figure 7 above (in magenta) with the light fixture keep out zone shown (in red) regardless if the fixtures are parallel or perpendicular to the TCC / TCIS. The cells are 1130.3mm by 1130.3mm (refer to Solutions layouts for exact central cell locations to coordinate lighting fixture locations).

2.1.3.2 Ambient Lighting – AR Field Facing Side of the TCC / TCIS

HARDWARE ENGINEERING REQUIREMENT: Deployment engineering shall validate that the specialized dark zone for the TCC / TCIS is achieved by following the measurement instructions of Section 5.1(4).

2.2 Lighting Requirements – Amazon Robotics Maintenance Cages

CONSTRUCTION REQUIREMENT: It is recommended that the maintenance cages **should** use dimmable lighting.

Note: Since maintenance cages may be located adjacent to stations, the use of dimmable lights will allow illuminance adjustments in order to meet individual station illuminance requirements.

2.3 Infrared Intensity

The drive units incorporate sensors that rely on infrared light for detecting obstacles. Therefore, it is important to ensure external infrared light sources are maintained to prevent interference. Below are the maximum infrared light radiation requirements to enable proper operation of the system.

- **Note:** Sunlight is a common source of IR radiation that may cause interference with drive units.

2.3.1 Infrared Intensity – All AR Drive Units

CONSTRUCTION REQUIREMENT: The intensity of infrared radiation between 700 nm and 1000 nm **shall** be less than $1.6e-5$ W/cm² at any given point on the Amazon Robotic floor on which the drive unit is operating.

- **Note:** This requirement prevents optical interference with the Drive Unit obstacle detection system. Violation of this requirement is usually caused by intense light sources or nearby windows allowing direct sunlight to the AR floor. See section 5.2 for instructions on how to perform these measurements.

3 Infrared Light Sources

3.1 Infrared Light Sources - Internal to AR

The following AR equipment includes an infrared light sources with the following wavelengths.

System	Light Source	Wavelength
H-DU	Obstacle Detection System	850 nm
P-L-DU	Obstacle Detection System	850 nm
P-DU	Obstacle Detection System	850 nm
P-T-DU	Obstacle Detection System	850 nm
S-DU / G-DU	Obstacle Detection System	808 nm
T-DU	Obstacle Detection System	850 nm

X-DU ¹	Obstacle Detection System	850 nm
Nike IDS	Intent Detection System	850 nm
FiDO	Light Curtain	850 nm
Proteus	Ouster OS0 LIDAR	865 nm
Proteus	D457 RealSense Stereo Camera	840 – 860 nm
Proteus	SICK TIM781 LIDAR	850 nm
Proteus	Charger IRDA Module	850 – 900 nm

3.1.1 Proteus Device Interference

There shall be no direct glare of 850nm – 900nm wavelength light where Proteus is operational. Direct glare is defined as any beam of light that enters uninterrupted (e.g. at the same wavelength and frequency as the beam that was emitted at the source) into the operational area of the Proteus drive unit. This requirement shall be in the purview of Robotic Deployment Engineering to ensure Proteus has a satisfactory lighting environment within the operational area of Proteus.

3.2 Infrared Light Sources – External to AR

CONSTRUCTION REQUIREMENT: Facilities constructed with external windows or rooftop skylights that allow sunlight to penetrate directly onto or in close proximity to an AR floor, pose a risk to violating the requirements described in section 2.3.1, as does any third-party supplied equipment that emits intense light.

Any facility constructed with external windows or rooftop skylights adjacent to an AR floor should be carefully evaluated, and if necessary, provided with an appropriate infrared filtering solution that fully satisfies the requirements in section 2.3.1, year round. Additionally, to determine compliance with the requirements in section 2.3.1, all buildings should be measured under worst case conditions appropriate to the time of year (i.e. low sun on the horizon through a window or high sun through a skylight) and taking into consideration the orientation of the building.

4 Aether RSP level Lighting Requirements

4.1 Aether System Overview

Aether is an AR controlled lighting panel that turns the AR field of lights ON or OFF across all AR field lights per RSP level. When personnel are not present on the AR field, the entire AR field will be turned OFF. When personnel activate a SRBRS vest and enter the field, the entire AR field of ceiling lighting will turn ON. Aether is a fail-safe system that will automatically turn the lights ON during a fire alarm event, Aether panel

¹The X-DU (Xanthus) obstacle detection system is available as both a passive or active stereo camera. The active stereo camera contains an illumination board that projects a pseudo-random dot pattern at 850nm. The passive version does not contain an illumination board.

power loss, or Aether panel malfunction. The Aether panel will be connected to the Main Safety Controller (MSC) making Aether an integrated part of the Amazon Robotics Floor Safety System (FSS). Building emergency lighting must not be controlled by the Aether panel.

4.2 Aether Construction Requirements

CONSTRUCTION REQUIREMENT: For buildings with AR enabled fulfillment systems to be compatible with Aether, the building ceiling lighting shall be capable of being controlled by a single switch. This shall allow all internal AR field ceiling lighting fixtures to be turned ON or OFF on a single switch without involving lighting fixtures outside the AR field including the green mile, storage areas outside the green mile on the RSP levels, and MHE occupied spaces.

Functional lighting requirements for AR enabled systems with dark zones or specialized lighting shall remain such that when Aether turns the AR field of lights ON, the dark zones remain intact for normal AR Station and TCC/TCIS operations. See figure 8 below. Throughout the AR field when the AR field building ceiling lighting is ON the lighting level shall be at 5-foot candles (50lux) unless otherwise specified in this document.

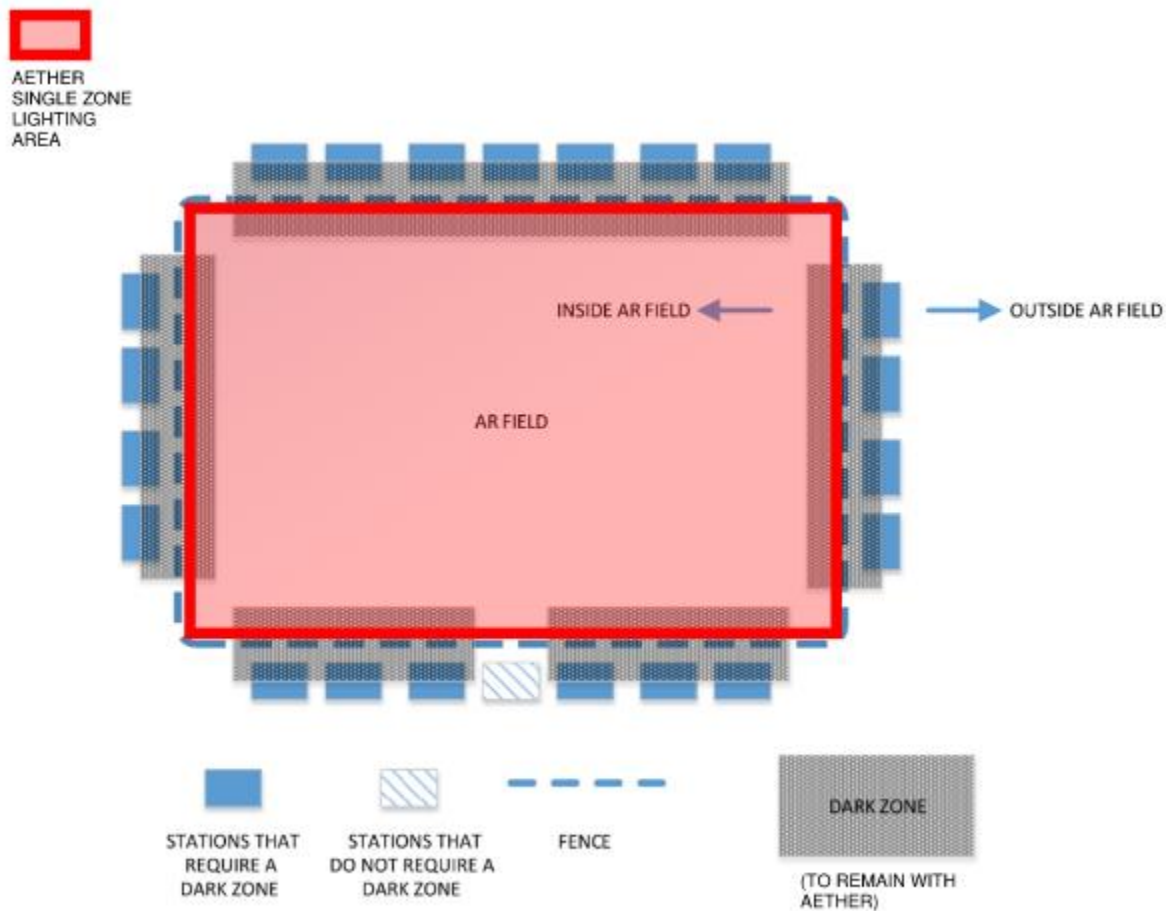


Figure 8 – Aether Ceiling Lighting Control Scope for a Fabric Pod Fulfillment RSP level Example

Building emergency lighting shall not be part of the Aether system and shall not be affected by Aether. Ceiling light fixtures that straddle over the AR fence line in the AR field and in the pedestrian area shall be omitted from the Aether single lighting zone to ensure proper lighting is maintained at all times in the green mile. Robotic Work cells outside the AR fence line shall be omitted from the Aether single lighting zone. Refer to ARPN# 811-00252 for the installation, operation, and maintenance manual FSS lighting control panel for building integration and system operation.

5 APPENDIX

5.1 Illuminance Measurement Procedure

1. Test equipment required:
 - a. Light Meter (Extech PN: LT300 or equivalent) – Measurement shall be taken with this light meter with the units set to “Fc” (foot-candles)

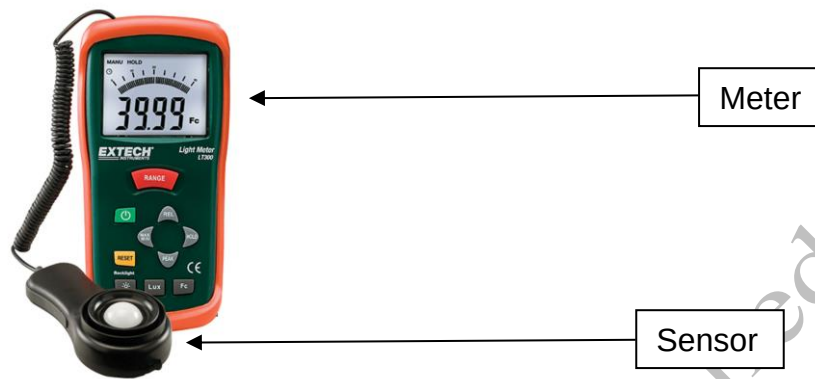


Figure 9 – Extech LT300 Light Meter

2. Station - Pod Face Illuminance Measurement (Hardware Engineering Measurement)
 - a. When the pod is present, position the light sensor against the pod fabric with the sensor pointing outwards (away from the pod). See figure 10 for six measurement locations - ensure the measurements approximately in these six locations comply with the requirement in section 2.1.2.1.
3. Station - Associate Work Area Measurement (Hardware Engineering Measurement)
 - a. Figure 11 depicts the Associate Work Area Zone.
 - b. Figure 12 depicts a top-down view of the associate work area with 6 areas where light measurements will be taken from a height of 915mm.
 - c. Position the light sensor approximately within each area noted in figure 12 (locations 7-12) at a height approximately 915mm from the finished floor with the sensor pointing upwards (towards the ceiling)
 - i. **Note:** Ensure there are no shadows covering the sensor from the person performing the measurements.
 - d. Perform measurement and record the resting measurement in each of the six locations. Note: resting measurement is defined as capturing a stable measurement from a fixed position which may take several seconds for the instrument to settle.
 - e. Log the measurements in a table similar figure 13 by measurement number.
 - f. Ensure the average value falls within the range as specified in section 2.1.1.1 and section 2.1.2.2 for stations.
 - g. Ensure the ratio between the Max value and Min value does not exceed 4:1.



Figure 10 – Pod Face Measurement Locations

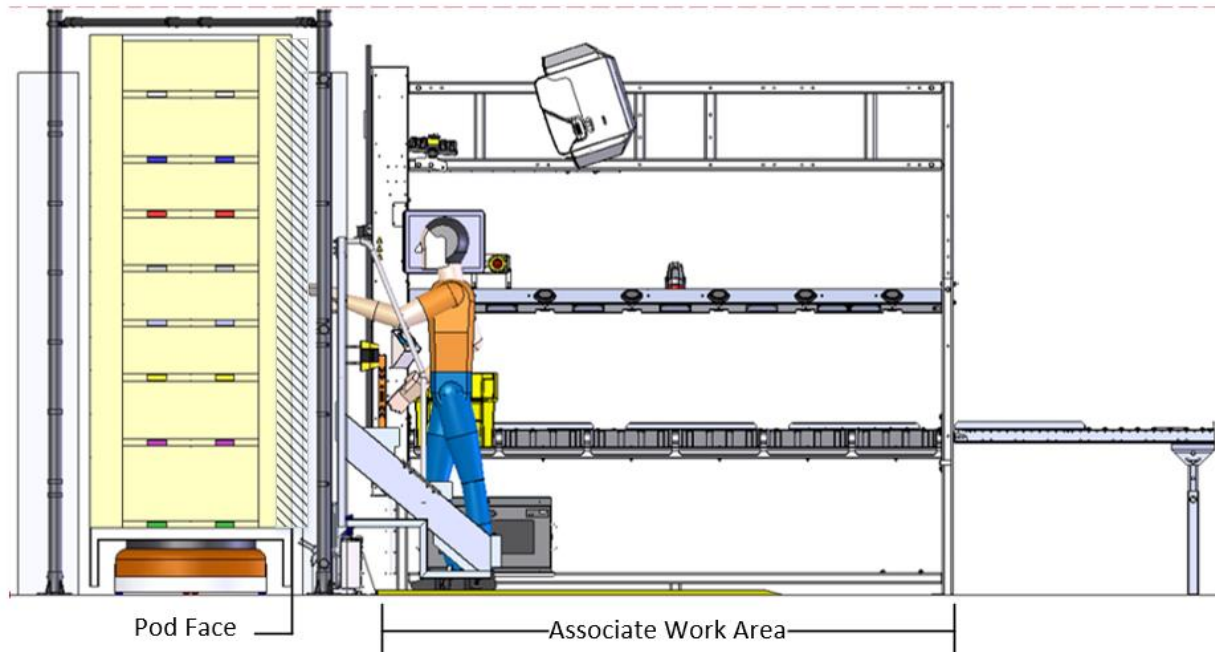


Figure 11 – Pod Face & Associate Work Area Locations

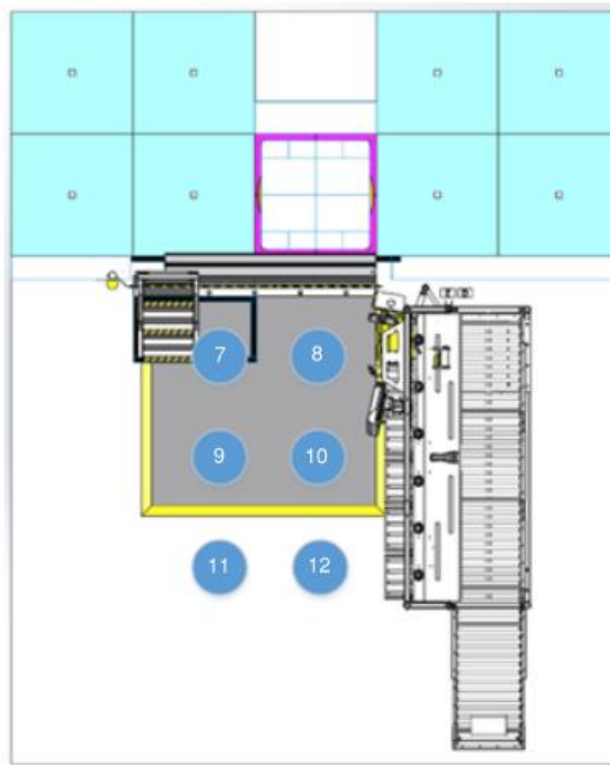


Figure 12 – Associate Work Area Approximate Measurement Locations

Measurement Location	Illuminance (Foot-Candles)
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
Sum of Measurements	
Average of Measurements (Sum of Measurements / 12)	
Min Value Measured	
Max Value Measured	
Ratio of Max to Min	

Figure 13 – Example Blank Measurement Table

4. TCC / TCIS specialized dark zone area (Hardware Engineering measurements)
 - a. Position the light sensor at the locations depicted in figure 14 below. The locations are where the face of the Hercules Dense Tote Pod meet the AR field side of the TCC / TCIS. Location (1, is center of fiducial cell page North), Location (2, is center of central fiducial cell), and Location (3, is center of fiducial cell page South) per figure 14 locations (1), (2), and (3) respectively.
 - b. Take measurements at 2650mm above the finished floor. Record the measurements using figure 15 as a reference.
 - c. The illuminance recorded shall be less than or equal to 35-foot candles (377lx). If the measurements are within specification at 2650mm above the finished floor then it is implied that the specialized dark zone is compliant for AR functionality closer to the finished floor.
 - d. Ensure no light fixtures are within 1000mm of the central cell (cell of location (2) in figure 14) in any direction of the edges of the cell as the bulbs of the fixture can cause reflective interference with the TCC / TCIS sensors.

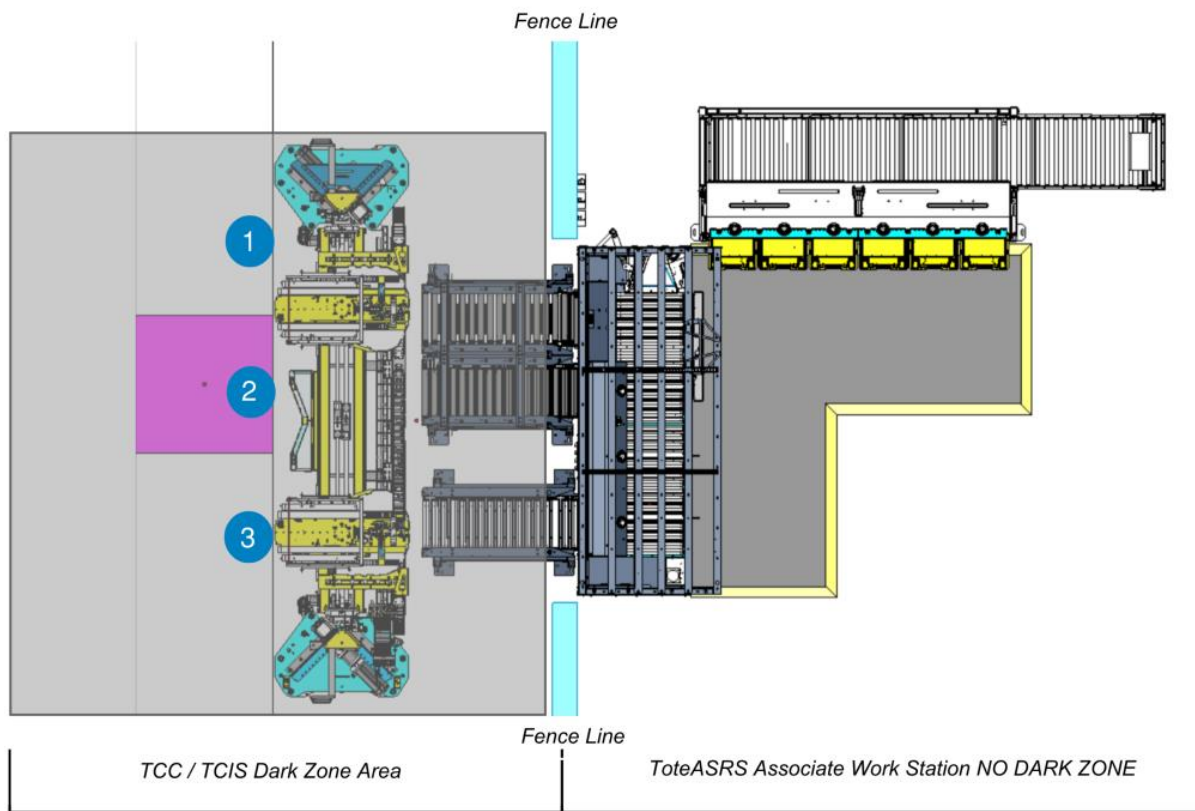


Figure 14 – TCC / TCIS Lighting Level Measurement Locations

Measurement Location	Illuminance (Foot-Candles)
1	
2	
3	

Figure 15 – Example Blank Measurement Table for the TCC / TCIS

Definition of Terms:

- **Sum of Measurements** - Add all the measurement location illuminance values together.
- **Average of Measurements** - Take the sum of measurements and divide by number of measurements.
- **Min Value** - Defined as the lowest measured illuminance value.
- **Max Value** - Defined as the highest measured illuminance value.
- **Ratio** - Divide the Max value by the Min value to yield the ration.
- **Resting Measurement** – To capture a stable measurement from a fixed point which may take several seconds to stabilize.

5.2 Infrared Radiation Measurement

1. Test equipment required:
 - a. Optical Meter (International Light Technologies PN: ILT2400 or similar) – Measurement shall be taken with this optical meter with the units set to “W/cm²”. See figure 16.
 - b. Broadband Silicon Detector (International Light Technologies PN: SED033. See figure 17.



Figure 16 – ILT2400 Optical Meter



Figure 17 – SED033 Silicone Detector

2. Measurement

- a. Position the detector on the floor where you intend to measure infrared radiation. Note: this could be in an area of the floor where sunlight is coming through a window or a skylight. See figure 18 for example.
- b. For all drive unit types, ensure the meter measurements comply with the requirement in section 2.3.1.



Figure 18 – Infrared Radiation Measurement on AR Floor